Name: **Enrolment No:**	UPES UNIVERSITY OF TOMORROW

UPES

End Semester Examination, May 2025

Course: Data Communication and Networks **Semester: IV**
Program: B.Tech-CSE/BCA (All Specializations) **Time: 03 hrs.**
Course Code: CSEG2065 **Max. Marks: 100**

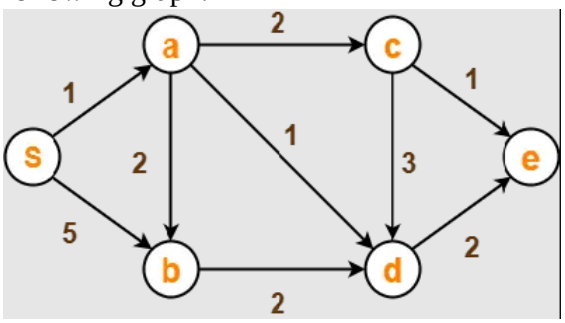
Instructions: a) Attempt all questions.
b) Scientific Calculator is allowed.

SECTION A
(5Qx4M=20Marks)

S. No.		**Marks**	**CO**
Q 1	Discuss various transmission impairments.	**4**	**CO1**
Q2	Draw Ipv4 header format and mention all fields.	**4**	**CO2**
Q3	Discuss the features of Wireshark.	**4**	**CO5**
Q4	Differentiate between port address and socket address with example.	**4**	**CO4**
Q5	Differentiate between hidden node and exposed node problem.	**4**	**CO4**

SECTION B
(4Qx10M= 40 Marks)

Q6	For the following ip addresses; 1.2.3.4 10.15.20.60 130.1.2.3 150.0.150.150 200.1.10.100 Identify the Class, Network IP Address, Direct broadcast address and Network ID of each IP Address.	**2+2+2+2+2**	**CO2**
Q7	Illustrate the concept of encapsulation and tunneling.	**2+8**	**CO3**
Q8	Discuss the concept of window-based flow control. Explain any one sliding window protocol with example.	**3+7**	**CO1**
Q9	Discuss the following congestion control strategies with suitable example: a) Back Pressure b) Choke Packet OR Elaborate the process of TCP connection establishment and termination. Support your explanation using neatly drawn diagrams.	**5+5** **OR** **5+5**	**CO3**

SECTION-C (2Qx20M=40 Marks)			
Q10	a) Discuss the Optimality principle. Use Dijkstra's Algorithm, find the shortest distance from source node 'S' to remaining vertices in the following graph:  b) Explain the concept of count to infinity problem in routing with example.	**4+10+6**	**CO3**
Q11	Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of slow start phase is 2 MSS and the threshold at the start of first transmission is 8 MSS. Assume that a time out occurs during the fifth transmission. Find the congestion window size at the end of tenth transmission. **OR** Differentiate between Subnetting and Supernetting with example. Suppose, We have a big single network having IP Address 200.1.2.0. We want to do subnetting and divide this network into 4 subnets. Find out the following for each subnet: • IP Address of the subnet • Total number of hosts that can be configured • Range of IP Addresses • Direct Broadcast Address	**10+10** **OR** **4+4+4+4+4**	**CO5**